Dr. Shengcheng Yu

shengcheng.yu@tum.de **yusc@nju.edu.cn** **https://seysc.com**



Postdoctoral Researcher at the Chair of Software Engineering & AI, School of Computation, Information and Technology, Technical University of Munich (TUM), collaborating with Prof. Dr. Chunyang Chen. B.Eng. and Ph.D. degrees in Software Engineering obtained from Nanjing University in 2020 and 2024, supervised by Prof. Zhenyu Chen and Assoc. Prof. Chunrong Fang. Research interests span intelligent software testing, automated GUI testing, crowdsourced testing, GUI automation, human-agent collaboration, and LLM personification and personalization for software engineering.

Experience

Technical University of Munich **School of Computation, Information and Technology**	*Postdoctoral Researcher*	Feb. 2025 – Now
Nanjing University **Software Institute**	*Research Assistant*	Oct. 2024 – Feb. 2025
Nanjing University **Software Institute**	*Ph.D.* *in Software Engineering*	Sep. 2020 – Sep. 2024
ETH Zurich **Department of Computer Science**	*Visiting Ph.D.* *in Software Engineering*	Oct. 2022 – Oct. 2023
Nanjing University **Software Institute**	*B.Eng.* *in Software Engineering*	Sep. 2016 – Jun. 2020

Publication (#: Co-first Author *: Corresponding Author)

Google Scholar Metrics: **Citations:1010** **h-index:16** **i10-index:22**

@: as (co-)first author / corresponding author

When Faster Decisions Do Not Make Better Agents: An Empirical Study of Jev in Agent Workflows (Preprint)
Yuchen Ling, **Shengcheng Yu***, Zhenyu Chen, Chunrong Fang*.

Security of Agent-Integrated Software: When Human Operations and Agent Actions Coexist (Preprint)
Ding Yang, Yuchen Ling, **Shengcheng Yu**, Zhenyu Chen, Chunrong Fang.

Agent-Integrated Software: Interaction Contracts and Continuous Assurance (Preprint)
Shengcheng Yu, Chunrong Fang*, Zhenyu Chen.

Software Engineering for and with GUI Agent (Preprint)
Shengcheng Yu, Yuchen Ling, Junyang Xing, Quan Zhou, Chunrong Fang*, Zhenyu Chen.

Toward Secure LLM Agents: Threat Surfaces, Attacks, Defenses, and Evaluation (Preprint)
Yuchen Ling, **Shengcheng Yu***, Zhenyu Chen, Chunrong Fang*.

Mobile App Test Requirement Generation via Multimodal Information Fusion
Peng Yang, Dayu Yang, **Shengcheng Yu**, Yunfeng Zhu, Yong Tang*.
[CCF-B] Journal of Systems and Software (JSS) 2026

Bridging Stakeholder and Product Requirements: An Empirical Study of Requirement Engineering in the Automotive Industry
Zixu Wang, **Shengcheng Yu***, Zhenchang Xing, Tobias Wenzel, Chunyang Chen.
[CCF-A] IEEE/ACM International Conference on Automated Software Engineering (ASE) 2026 - Industry Showcase Track

More than a Judge: An Empirical Study of Agent-Human Interaction in Crowdsourced Testing Assessment
Yue Wang, Yuan Zhao*, **Shengcheng Yu**, Zhenyu Chen, Qing Gu.
[CCF-A] ACM Transactions on Software Engineering and Methodology (TOSEM) 2026

Can Trajectory-Level Verdicts Be Trusted? An Empirical Study of Post-hoc Trajectory Interpretation for Mobile GUI Agents (Accepted)
Yunfeng Zhu, **Shengcheng Yu**, Zhenyu Chen*.
[CCF-C] IEEE International Conference on Software Quality, Reliability, and Security (QRS) 2026

Scenario-Guided LLM-based Mobile App GUI Testing
Shengcheng Yu, Yuchen Ling, Chunrong Fang*, Quan Zhou, Yi Zhao, Chunyang Chen, Shaomin Zhu, Zhenyu Chen.
[CCF-A] ACM Transactions on Software Engineering and Methodology (TOSEM) 2026. *Accepted to present at ASE 2026 Journal-First Track*

Towards Automated Crowdsourced Testing via Personified-LLM
Shengcheng Yu, Yuchen Ling, Chunrong Fang, Zhenyu Chen, Chunyang Chen.
[CCF-A] ACM International Conference on the Foundations of Software Engineering (FSE) 2026

Large Language Models for Software Testing Education: an Experience Report
Peng Yang, Yunfeng Zhu, Chao Chang, **Shengcheng Yu**, Zhenyu Chen, Yong Tang*.
[CCF-A] ACM International Conference on the Foundations of Software Engineering (FSE) 2026 - Software Engineering Education Track

Software Testing Education Innovation for the Domestic Software and Hardware Ecosystem
Zhenyu Chen, Chunrong Fang, **Shengcheng Yu**, Jiawei Liu.
Computer Education (in Chinese) 2026

Survey on Application of Retrieval-augmented Generation in Software Engineering
Quanjun Zhang, Chunrong Fang*, **Shengcheng Yu**, Yuan Zhao, Zhenyu Chen.
[CCF-A (Chinese)] Journal of Software (in Chinese) 2026

GUI Test Migration via LLM with Scenario-Granularity Understanding
Quan Zhou, **Shengcheng Yu**, Chunrong Fang*, Junyang Xing, Hongyi Yang, Jia Liu, Wenbin Guo, Zhenyu Chen.
[CCF-B] Frontiers of Computer Science (FCS) 2026 - Letter

A Survey on Large Language Models for Software Engineering
Quanjun Zhang, Chunrong Fang, Yang Xie, Yaxin Zhang, **Shengcheng Yu**, Weisong Sun, Yun Yang, Zhenyu Chen.
[CCF-A] Science China Information Sciences (SCIS) 2026

Human Cognitive Pattern Simulation for Crowdsourced Test Report Consistency Detection
Yuchen Ling, **Shengcheng Yu**, Shuguang Chen, Liuming Wang, Chunrong Fang*, Wenbin Guo, Jia Liu, Zhenyu Chen.
[CCF-A] IEEE Transactions on Software Engineering (TSE) 2026

Vision-Based Mobile App GUI Testing: A Survey
Shengcheng Yu, Chunrong Fang*, Ziyuan Tuo, Quanjun Zhang, Chunyang Chen, Zhenyu Chen, Zhendong Su.
ACM Computing Surveys (CSUR) 2025

TestBench: Evaluating Class-Level Test Case Generation Capability of Large Language Models
Quanjun Zhang, Ye Shang, Chunrong Fang*, Siqi Gu, **Shengcheng Yu**, Jianyi Zhou, Zhenyu Chen*.
[CCF-B] Frontiers of Computer Science (FCS) 2025

Multi-dimensional Assessment of Crowdsourced Testing Reports via LLMs
Yue Wang, Yuan Zhao*, **Shengcheng Yu**, Zhenyu Chen.
[CCF-A] IEEE/ACM International Conference on Automated Software Engineering (ASE) 2025

LLM-based Crowdsourced Test Report Clustering
Yuchen Ling, **Shengcheng Yu**, Chunrong Fang*, Quan Zhou, Zhenyu Chen.
[CCF-A] ACM Transactions on Software Engineering and Methodology (TOSEM) 2025

Text–Image Fusion Template for Large Language Model Assisted Crowdsourcing Test Aggregation
Yunfeng Zhu, **Shengcheng Yu**, Zhaowei Zong, Yue Wang, Yuan Zhao*, Zhenyu Chen.
[CCF-B] Journal of Systems and Software (JSS) 2025

Test Script Intention Generation for Mobile Application via GUI Image and Code Understanding
Shengcheng Yu, Chunrong Fang*, Jia Liu*, Zhenyu Chen.
[CCF-A] ACM Transactions on Software Engineering and Methodology (TOSEM) 2025

Semantic Matching-based Cross-Platform Mobile App Test Script Record and Replay via Large Language Models
Shengcheng Yu, Chunrong Fang*, Ye Zhong, Quanjun Zhang, Qin Liu, Jia Liu, Tao Zheng, Zhenyu Chen.
[CCF-A (Chinese)] Journal of Software (in Chinese) 2025

Improving Retrieval-Augmented Deep Assertion Generation via Joint Training
Quanjun Zhang, Chunrong Fang*, Yi Zheng, Ruixiang Qian, **Shengcheng Yu**, Yuan Zhao, Jianyi Zhou, Yun Yang, Tao Zheng, Zhenyu Chen.
[CCF-A] IEEE Transactions on Software Engineering (TSE) 2025

Enhanced Crowdsourced Test Report Prioritization via Image-and-Text Semantic Understanding and Feature Integration
Chunrong Fang, **Shengcheng Yu***, Quanjun Zhang, Xin Li, Yulei Liu, Zhenyu Chen.
[CCF-A] IEEE Transactions on Software Engineering (TSE) 2025

Redefining Crowdsourced Test Report Prioritization: An Innovative Approach with Large Language Model
Yuchen Ling, **Shengcheng Yu**, Chunrong Fang*, Guobin Pan, Jun Wang, Jia Liu*.
[CCF-B] Information and Software Technology (IST) 2024

Practical, Automated Scenario-based Mobile App Testing
Shengcheng Yu, Chunrong Fang*, Mingzhe Du, Zimin Ding, Zhenyu Chen, Zhendong Su.
[CCF-A] IEEE Transactions on Software Engineering (TSE) 2024

Effective, Platform-Independent GUI Testing via Image Embedding and Reinforcement Learning
Shengcheng Yu, Chunrong Fang*, Xin Li, Yuchen Ling, Zhenyu Chen, Zhendong Su.
[CCF-A] ACM Transactions on Software Engineering and Methodology (TOSEM) 2024

Semi-Supervised Crowdsourced Test Report Clustering via Screenshot-Text Binding Rules
Shengcheng Yu, Chunrong Fang*, Quanjun Zhang, Mingzhe Du, Jia Liu*, Zhenyu Chen*.
[CCF-A] ACM International Conference on the Foundations of Software Engineering (FSE) 2024

Practical Non-Intrusive GUI Exploration Testing with Visual-based Robotic Arms
Shengcheng Yu, Chunrong Fang*, Mingzhe Du, Yuchen Ling, Zhenyu Chen, Zhendong Su.
[CCF-A] IEEE/ACM International Conference on Software Engineering (ICSE) 2024

LLM for Test Script Generation and Migration: Challenges, Capabilities, and Opportunities
Shengcheng Yu, Chunrong Fang*, Yuchen Ling, Chentian Wu, Zhenyu Chen.
[CCF-C] IEEE International Conference on Software Quality, Reliability, and Security (QRS) 2023

Towards Effective Bug Reproduction for Mobile Applications
Xin Li#, **Shengcheng Yu#**, Lifan Sun, Yuexiao Liu, Chunrong Fang*.
International Conference on Dependable Systems and Their Applications (DSA) 2023. ***Best Paper Award***

Mobile App Crowdsourced Test Report Consistency Detection via Deep Image-and-Text Fusion Understanding
Shengcheng Yu, Chunrong Fang*, Quanjun Zhang, Zhihao Cao, Yexiao Yun, Zhenfei Cao, Kai Mei, Zhenyu Chen.
[CCF-A] IEEE Transactions on Software Engineering (TSE) 2023

Test Report Generation for Android App Testing via Heterogeneous Data Analysis
Chunrong Fang#*, **Shengcheng Yu#**, Ting Su*, Jing Zhang, Yuanhan Tian, Yang Liu (Co-first Author).
[CCF-A] IEEE Transactions on Software Engineering (TSE) 2023

Leveraging Android Automated Testing to Assist Crowdsourced Testing
Xiuting Ge, **Shengcheng Yu**, Chunrong Fang*, Qi Zhu, Zhihong Zhao.
[CCF-A] IEEE Transactions on Software Engineering (TSE) 2023

SemCluster: A Semi-Supervised Clustering Tool for Crowdsourced Test Reports with Deep Image Understanding
Mingzhe Du, **Shengcheng Yu**, Chunrong Fang*, Tongyu Li, Heyuan Zhang, Zhenyu Chen.
[CCF-A] ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering (ESEC/FSE) 2022 - Demonstrations Track

Test Case Prioritization Using Partial Attention
Quanjun Zhang, Chunrong Fang[*], Weisong Sun, **Shengcheng Yu**, Yutao Xu, Yulei Liu.
[CCF-B] Journal of Systems and Software (JSS) 2022

UniRLTest: Universal Platform-Independent Testing with Reinforcement Learning via Image Understanding
Ziqian Zhang, Yulei Liu, **Shengcheng Yu**, Xin Li, Yexiao Yun, Chunrong Fang[*], Zhenyu Chen.
[CCF-A] ACM SIGSOFT International Symposium on Software Testing and Analysis (ISSTA) 2022 - Tool Demonstrations Track

Layout and Image Recognition Driving Cross-platform Automated Mobile Testing
Shengcheng Yu, Chunrong Fang[*], Yexiao Yun, Yang Feng.
[CCF-A] IEEE/ACM International Conference on Software Engineering (ICSE) 2021

Prioritize Crowdsourced Test Reports via Deep Screenshot Understanding
Shengcheng Yu, Chunrong Fang[*], Zhenfei Cao, Xu Wang, Tongyu Li, Zhenyu Chen.
[CCF-A] IEEE/ACM International Conference on Software Engineering (ICSE) 2021

STIFA: Crowdsourced Mobile Testing Report Selection Based on Text and Image Fusion Analysis
Zhenfei Cao, Xu Wang, **Shengcheng Yu**, Yexiao Yun, Chunrong Fang[*].
[CCF-A] IEEE/ACM International Conference on Automated Software Engineering (ASE) 2020 - Tool Demonstrations Track

FuRong: Fusing Report of Automated Android Testing on Multi-Devices
Yuanhan Tian[#], **Shengcheng Yu**[#], Chunrong Fang[*], Peiyuan Li.
[CCF-A] IEEE/ACM International Conference on Software Engineering (ICSE) 2020 - Demonstrations Track

LIRAT: Layout and Image Recognition Driving Automated Mobile Testing of Cross-Platform
Shengcheng Yu, Chunrong Fang[*], Yang Feng, Wenyuan Zhao, Zhenyu Chen.
[CCF-A] IEEE/ACM International Conference on Automated Software Engineering (ASE) 2019 - Demonstrations Track

Crowdsourced Report Generation via Bug Screenshot Understanding
Shengcheng Yu.
[CCF-A] IEEE/ACM International Conference on Automated Software Engineering (ASE) 2019 - Student Research Competition Track. ***First Place Award (Undergraduate) Winner***

From Data Quality to Model Quality: an Exploratory Study on Deep Learning
Tianxing He, **Shengcheng Yu**, Ziyuan Wang[*], Jieqiong Li, Zhenyu Chen[*].
[CCF-C] Asia-Pacific Symposium on Internetware (Internetware) 2019

An Exploratory Study on Judicial Image Quality Assessment Based on Deep Learning
Qiqi Gu, Weilin Cai, **Shengcheng Yu**, Zhenyu Chen.
[CCF-C] IEEE International Conference on Software Quality, Reliability, and Security (QRS) 2019

Award

- ASE 2026 Distinguished Reviewer Award
- 2026 Nanjing University Outstanding Ph.D. Dissertation
- 2025 ACM Certified Reviewer
- 2023 Nanjing University Dongliang Scholarship (Award of Excellence in Academic Achievements)
- 2023 Nanjing University Excellent Graduate Student Pacemaker
- 2023 Nanjing University Excellent Graduate Student Innovation Project
- 2021 National Scholarship for Graduate Students
- 2021 Nanjing University Excellent Graduate Student Pacemaker
- 2021 College Student Innovation Annual Conference Elected Project of Ministry of Education
- 2020 Nanjing University - HPI Research Scholarship
- 2020 Nanjing University President's Special Scholarship for Ph.D. Student
- 2020 Nanjing University Outstanding Undergraduate Thesis - First Prize
- 2020 Nanjing University Software Institute Outstanding Graduate
- 2019 ACM SRC First Place Winner of Undergraduate Group in ASE 2019
- 2019 ACM SIGSOFT CAPS Program Award

Funding

Principal Investigator: Deutsche Forschungsgemeinschaft (DFG, German Research Foundation) (583997361)
KoQaGUI: Knowledge-Guided Quality Assurance of Software GUI

Co-Principal Investigator: ACM SIGSOFT Funding for Summer / Winter School
LLM Software Development

Key Participant: National Key R&D Program of China (2024YFF0908000)
Technology Research, Development, and Application of Open Source and Open Innovation Service Platform

Principal Investigator: Postgraduate Research & Practice Innovation Program of Jiangsu Province (KYCX22_0174)
Crowdsourced Test Report Optimization based on Multi-Modal Information Fusion Understanding

Key Participant: National Natural Science Foundation of China (62141215)
Crowdsourced Testing for Ubiquitous Operating System (UOS) Open Source Ecosystem

Principal Investigator: Nanjing University Training Program of Innovation for Undergraduates (202010284073Z)
Mobile Application Testing Research Based on Image Understanding

Participant: National Key R&D Program of China (2018YFB1403400)
R&D and Application of Integrated Crowdsourcing Test Service Platform for Information Products and Technology Services

Participant: National Natural Science Foundation of China (61802171)
Human-Machine Collaborative Mobile Application Testing Based on Comprehensible Information Fusion

Teaching

- **TUM: Seminar Automated Mobile App Testing (IN0014, IN2107, IN45110, INHN0015/ INHN4010):** Lecturer (Summer / Winter 2025, 2026, 2027)
- **TUM: Development of LLM-driven GUI Agents (IN2106):** Lecturer (Summer 2025)
- **NJU: Automated Testing:** Teaching Assistant (2019, 2020, 2021, 2023, 2024)
- **NJU: Software Engineering and Computing III:** Teaching Assistant (2021, 2023)
- **NJU: Foundation of Computing System:** Teaching Assistant (2021)

Academic Service

- **Conference Organizing Committee:**
  - LLM-Driven Software Testing Workshop (with ChinaSoft) (2026): Co-Chair
  - UISE Workshop (with ICSE) (2026): Web & Publicity Chair
- **Journal Reviewer:**
  - ACM Transactions on Software Engineering and Methodology (TOSEM)
  - IEEE Transactions on Software Engineering (TSE)
  - IEEE Transactions on Dependable and Secure Computing (TDSC)
  - Journal of Systems & Software (JSS)
  - Information and Software Technology (IST)
  - Journal of Software: Evolution and Process
  - Empirical Software Engineering (EMSE)
  - International Journal of Human-Computer Interaction (IJHCI)
  - IEEE Transactions on Reliability (TRel)
  - IEEE Transactions on Intelligent Vehicles (TIV)

- **Conference Technical/Research Track PC Member:**
  - International Conference on Software Engineering (ICSE): 2027 (also Shadow PC Area Chair)
  - International Conference on the Foundations of Software Engineering (FSE): 2027
  - IEEE/ACM International Conference on Automated Software Engineering (ASE): 2025, 2026 (also ACM SRC Judge and Session Chair)
  - ACM SIGSOFT International Symposium on Software Testing and Analysis (ISSTA): 2026
  - International Joint Conferences on Artificial Intelligence (IJCAI): 2025, 2026
  - IEEE International Conference on Software Maintenance and Evolution (ICSME): 2026
  - ACM Conference on Intelligent User Interfaces (IUI): 2025, 2026, 2027
  - Mining Software Repositories Conference (MSR): 2024, 2027
  - International Workshop on Advances in Mobile App Analysis (A-Mobile): 2025
- **Conference Artifact Evaluation Track PC Member:**
  - International Conference on Software Engineering (ICSE): 2024
  - ACM SIGSOFT International Symposium on Software Testing and Analysis (ISSTA): 2024, 2025

Invited Talk

- **2026-09-24 Oxford Internet Institute:** "Engineering Agent-Integrated Software: New Perspective of Human-Agent-Software Interaction"
- **2025-06-18 Fortiss @ Munich:** "Intelligent GUI Testing with Knowledge Enhancement & Persona-based AI"
- **2025-04-09 IPAI @ Heilbronn:** "Beyond Chatbot: GUI Automation by LLM-based Agents"

Last Updated: 2026-10-12